MATH 1342 – Elementary Statistics – Week 11: Normal Distributions and Inference for a Proportion

Nov 3 – 5 | 5.1 Hypothesis Testing Using Normal Distributions • 6.1 Inference for a Proportion (Distribution, CI, Hypothesis Test) | Fall 2026 · Section 036 · TTh 6:00–7:20 PM (DIL)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Explain when the Central Limit Theorem justifies replacing a simulated randomization distribution with a normal distribution N(null value, SE). (CLO 5) **2.** Compute a standardized z statistic and find the p-value as a right-tail, left-tail, or two-tail area of the standard normal distribution. (CLO 3, CLO 8) **3.** Check the large-sample condition (at least 10 in each category) for inference about one proportion. (CLO 5) **4.** Construct and interpret a confidence interval for a population proportion with z* for 90%, 95%, or 99% confidence. (CLO 6) **5.** Carry out a one-proportion z test and conclude in context, and explain why the CI standard error uses $\hat{p}$ while the test uses p_0. (CLO 8)	e-Text: 5.1 Hypothesis Tests Using Normal Distributions Interactive Lecture: 5.1 Hypothesis Tests Using Normal Distributions Slides: 5.1 Hypothesis Tests Using Normal Distributions e-Text: 6.1 Inference for a Proportion (6.1-D Distribution, 6.1-CI Confidence Interval, 6.1-HT Hypothesis Test) Interactive Lecture: 6.1 Inference for a Proportion Slides: 6.1 Inference for a Proportion StatKey – Theoretical Distributions (Normal)	Participation: Normal Distributions and z – malaria/mosquito example: compute z, shade the tail in StatKey, find the p-value Participation: Inference for a Proportion – 4-step CI and 4-step test for a proportion (voting example); compare the two SEs	Online Homework: 5.1 Hypothesis Tests Using Normal Distributions (due Sun Nov 8, 11:59 PM) Online Homework: 6.1 Inference for a Proportion (due Sun Nov 8, 11:59 PM)	WileyPLUS StatKey Blackboard Zoom (live class)
Course resources and assessments of earlier weeks (not aligned to this week's objectives)			Test 2 Retake – Mon Nov 2 – Sun Nov 8 (optional; higher score counts)	Assessment Tool in Blackboard Respondus LockDown Browser and Monitor

CLO 3 Compute and interpret empirical and theoretical probabilities using the rules of probabilities and combinatorics. **CLO 5** Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables. **CLO 6** Describe and compute confidence intervals. **CLO 8** Perform hypothesis testing using statistical methods.